

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Practical Lab Exam 2

Practical Lab Exam 2

Class: Class 10

Assessment Type: Practical Lab Exam 2

Total Marks: 20

Duration: 45 minutes

Topics Covered: SQL Database and Python with Database

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Lab Tasks (15 marks)

Task 1: SQL Queries (5 marks)

Write SQL queries for a **Bookstore Database** with the following tables:

- Books(book_id INT PRIMARY KEY, title VARCHAR(50), author VARCHAR(30), price DECIMAL, genre VARCHAR(20), stock INT)
- Customers(cust_id INT PRIMARY KEY, name VARCHAR(30), email VARCHAR(50), city VARCHAR(20))
- Sales(sale_id INT PRIMARY KEY, book_id INT, cust_id INT, quantity INT, sale_date DATE)

Write queries for: (a) List all books with price above 500 sorted by title (b) Find customers from 'Bangalore' or 'Mysore' (c) Count the number of books in each genre (d) Find the total revenue (price – quantity) for each book (e) Find the top 3 selling books by total quantity sold (f) Find customers who have not made any purchase (g) Display the monthly sales summary (month, total_sales, total_books_sold)

Task 2: Python with SQLite Database (5 marks)

Write a Python program that:

- Creates a SQLite database for a **Student Result System**
- Creates tables: Students(roll, name, class), Subjects(sub_id, sub_name, max_marks), Marks(roll, sub_id, marks)
- Inserts at least 8 student records with marks in 4 subjects
- Performs the following operations:
 1. Display all students with their marks
 2. Calculate and display each student's total and percentage
 3. Find the class topper
 4. Find subject-wise toppers
 5. List students who failed (percentage < 40%)
 6. Display the rank list
- Uses proper exception handling for database operations
- Closes the connection properly

Task 3: Python - Data Analysis with CSV (5 marks)

Write a Python program that:

- Reads a CSV file sales_data.csv with columns: Date, Product, Quantity, Price, Region
- Performs the following analyses:
 1. Total sales amount by region
 2. Best-selling product (by quantity)
 3. Daily sales trend (group by date)

4. Region with highest average order value
 5. Products with zero or negative stock
- Writes the analysis report to a new file `analysis_report.txt`
 - Uses try-except for file operations
 - Displays results in a formatted table

Viva Voce (5 marks)

1. What is a **primary key**? Can a table have composite primary key? Give an example. (1 mark)
2. Explain the difference between `INNER JOIN` and `LEFT JOIN` with diagrams. When would you use each? (1 mark)
3. What is the purpose of `GROUP BY` clause? How is it different from `ORDER BY`? (1 mark)
4. How does Python's `sqlite3` module handle database connections? What happens if you forget to `commit()` changes? (1 mark)
5. What is **normalization** in databases? Explain 1NF and 2NF with an example. (1 mark)